

Risk Plan – Inventory Management System

Ref	Identified Risk	Probability	Severity	Risk Management Strategy Comments (CO) / Mitigation (MT) / Amelioration (AM) / Opportunity (OP) / Fallback (FB)
1	MVP scope expansion, unclear acceptance criteria, and new stakeholder requests	High	High	MT: Establish strict change control. AM: Baseline MVP scope. FB: Defer non-critical features.
1	Limited team availability, under-estimation, and delayed reviews	High	High	MT: Improve sprint estimation and buffers. AM: Cross-train members. FB: Adjust scope
1	Serverless routing issues, Mongo transaction constraints, FEFO logic defects	Medium	High	Add integration and architecture validation. AM: Edge case testing. FB: Rollback deployment
1	RBAC bypass, token validation failures, and secrets leakage	Medium	High	MT: Enforce RBAC and token validation. AM: Secure secrets storage. FB: Disable affected modules
1	Incorrect stock movements, negative stock, and a missing audit trail	Medium	High	MT: Implement stock validation rules. AM: Enable audit trail. FB: Manual reconciliation
1	Low test coverage, defects not caught before demo/deploy	High	High	MT: Increase automated testing. AM: CI quality gates. FB: Hotfix patch cycle
1	CORS misconfiguration, Vercel route mapping 404s, and environment variable mistakes	Medium	Medium	MT: Validate CORS and routing configs. AM: Pre-deploy checklist. FB: Rapid redeploy
1	Firebase Auth downtime, MongoDB Atlas issues, Vercel incidents	Low	High	CO: Third-party dependency risk. MT: Monitoring alerts. FB: Switch to backup services if available
1	User training gaps, incorrect workflow usage, and poor documentation	Medium	Medium	MT: Provide structured training. AM: Improve documentation. FB: On-demand support sessions

Key: **CO**=comments to explain risk; **MT**=actions to reduce probability of risk; **AM**=safeguards to decrease severity; **OP**=positive outcomes; **FB**=back up (“Plan B”)